

A COUNTEREXAMPLE TO LITERAL STABILIZATION OF FC_α FOR BOREL FIELDS OF GROUPS

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ABSTRACT. Berendschot, Chakraborty, Donvil, Kim, and Klisse asked whether the transfinite sequence $FC_\alpha(\mathcal{G})$ associated with a discrete measured field of groups must eventually stabilize. We give a negative answer to this literal question. A Borel field of McLain groups over the standard Borel space of countable linear orders has fibers of unbounded countable FC-rank, so its global FC_α -sequence cannot stabilize. The non-stabilizing fibers may be placed on a null part of the unit space, so the modulo-null stabilization theorem in the source paper is essentially sharp.

1. SOURCE QUESTION

The source paper is T. Berendschot, S. Chakraborty, M. Donvil, S.-J. Kim, and M. Klisse, *The Choquet-Deny Property for Groupoids*, arXiv:2406.05004. Definition 2.12 defines, for a discrete Borel field of groups $\mathcal{G}$, a transfinite sequence $FC_\alpha(\mathcal{G})$ for $\alpha < \omega_1$. Proposition 2.13 proves that this sequence stabilizes outside a negligible set. Remark 2.15 asks whether the sequence itself must eventually stabilize.

is a null set. By Proposition 2.13 and Corollary 2.11, $(\mathcal{G}_\alpha)_x$ is an icc group for μ -almost every $x \in \mathcal{G}_\alpha^{(0)}$. By [BCDK24, Proposition 6.1], the discrete measured field of groups $(\mathcal{G}_\alpha, \mu)$ is icc if and only if μ -almost every $(\mathcal{G}_\alpha)_x$ is icc, hence completing the proof. $\square$

Remark 2.15. As remarked in the preamble to Proposition 2.13, we do not currently know whether $FC_\alpha(\mathcal{G})$ must eventually stabilize. It would be interesting to know whether this is true since this would allow us to define a definitive notion of the FC-hypercenter of a discrete measured field of groups.

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2. IDEA OF THE PROOF

The modulo-null statement cannot control what happens on a null Borel part of the unit space. McLain's classical construction gives countable groups of arbitrarily large countable FC-rank. These groups can be arranged uniformly as a Borel field by letting the base space be the standard Borel space of all countable linear orders and taking, over each order, the corresponding finitary McLain unitriangular group. Since the FC_α -recursion is

Date: 2026-07-19.

computed fiber by fiber for a field of groups, unbounded fiber ranks force the global sequence to keep gaining elements after every countable stage.

3. THE COUNTEREXAMPLE

Let LO be the standard Borel space of strict linear orders on $\mathbb{N}$. For $L \in \text{LO}$, write $i <_L j$ for the order relation. Let $M(L)$ be the McLain group over $\mathbb{F}_2$ attached to $(\mathbb{N}, L)$: an element is a finite $\mathbb{F}_2$ -valued function a on

$$P_L = \{(i, j) \in \mathbb{N}^2 : i <_L j\}.$$

Multiplication is the usual finitary unitriangular multiplication

$$(a * b)(i, j) = a(i, j) + b(i, j) + \sum_{i <_L k <_L j} a(i, k)b(k, j),$$

where the sum is finite because a and b have finite support. The zero function is the identity. Inversion is the usual finite unitriangular inverse, equivalently the finite geometric series for $(I + a)^{-1}$.

Lemma 1. *The family $(M(L))_{L \in \text{LO}}$ is a discrete Borel field of countable groups.*

Proof. Code finite $\mathbb{F}_2$ -valued functions on $\mathbb{N}^2$ by natural numbers. The total space

$$\mathcal{G} = \{(L, a) : L \in \text{LO}, \text{supp}(a) \subseteq P_L\}$$

is Borel in $\text{LO} \times \mathbb{N}$, because membership of each finite support pair in P_L is checked by finitely many coordinates of L . The source and target maps are both $(L, a) \mapsto L$, so every fiber is countable.

For fixed finite codes a, b, c , the relation $a * b = c$ is decided by finitely many statements of the form $i <_L j$ and finitely many additions in $\mathbb{F}_2$. Hence multiplication is Borel. The inverse is Borel for the same reason, since the subalgebra generated by a finite support element is finite dimensional and the unitriangular inverse is a finite expression in the same support closure. Thus $\mathcal{G}$ is a discrete Borel field of groups. $\square$

Equip LO with any probability measure μ , for instance a point mass at one fixed order. Since $\mathcal{G}$ is a field of groups, the source and target measures coincide; hence $(\mathcal{G}, \mu)$ is a discrete measured field of groups.

Lemma 2. *For every countable ordinal α and every $L \in \text{LO}$,*

$$FC_\alpha(\mathcal{G})_L = FC_\alpha(M(L)).$$

Proof. This is a direct induction on α . At $\alpha = 0$ both sides are the identity subgroup. If the equality holds at α , then the quotient field $\mathcal{G}/FC_\alpha(\mathcal{G})$ has fiber $M(L)/FC_\alpha(M(L))$ over L . The conjugacy class of an element in a field of groups is its ordinary conjugacy class inside the same fiber, so taking the FC-center of the quotient field is exactly the fiberwise operation. This proves the successor step. For a limit ordinal, both definitions take the union over earlier stages. $\square$

Theorem 1. *There exists a discrete measured field of countable groups $(\mathcal{G}, \mu)$ such that the sequence $FC_\alpha(\mathcal{G})$, $\alpha < \omega_1$, does not eventually stabilize.*

Proof. Use the field $(\mathcal{G}, \mu)$ constructed above. McLain's ordered-group construction realizes arbitrary infinite ordinal FC-ranks: for every infinite countable ordinal ξ , there is a countable linear order L_ξ such that $M(L_\xi)$ has FC-rank ξ . This is precisely the McLain fact cited in the source paper immediately before Definition 2.12.

Suppose, toward a contradiction, that $FC_\alpha(\mathcal{G})$ eventually stabilizes; say $FC_\beta(\mathcal{G}) = FC_\alpha(\mathcal{G})$ for every $\beta > \alpha$. Choose an infinite countable ordinal $\xi > \alpha$ and a McLain order L_ξ for which $M(L_\xi)$ has FC-rank ξ . Then

$$FC_\alpha(M(L_\xi)) \neq M(L_\xi) = FC_\xi(M(L_\xi)).$$

Pick $g \in M(L_\xi) \setminus FC_\alpha(M(L_\xi))$. By Lemma 2,

$$(L_\xi, g) \notin FC_\alpha(\mathcal{G}) \quad \text{but} \quad (L_\xi, g) \in FC_\xi(\mathcal{G}),$$

contradicting the assumed stabilization. Therefore the global FC_α -sequence does not eventually stabilize. $\square$

Remark 1. *The measure can be chosen to be a point mass on a single harmless fiber. Thus the failure of literal stabilization may occur entirely on a null part of the unit space. This is why Proposition 2.13 in the source, which proves stabilization modulo negligible sets, is not contradicted.*

4. NOVELTY AND LIMITATIONS

Local run indexes were searched for arXiv:2406.05004, FC_α , McLain groups, FC-rank, and Choquet-Deny groupoids. A bounded web search on 2026-07-19 for exact phrases around the stabilization question found the source paper and no later resolution. The proof depends on the standard McLain rank-realization theorem already cited by the source paper; the new step is packaging those countable McLain groups into one Borel field.

REFERENCES

- [1] T. Berendschot, S. Chakraborty, M. Donvil, S.-J. Kim, and M. Klisse, *The Choquet-Deny Property for Groupoids*, arXiv:2406.05004.
- [2] D. H. McLain, *Remarks on the upper central series of a group*, Glasgow Mathematical Journal 3 (1956), no. 1, 38–44.